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# Department of CSE

B.Tech. CSE Trimester -5

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# Cloud Infrastructure and Services

**25CS2206E**

**Skill**

**2026**

**Experiment -1**

**AIM: Create an IAM user with programmatic access only, download the access key and secret key, and document the steps with screenshots. Then set up a billing alert for the account so that an email is received when the Free Tier limit is about to be crossed.**

**Description:**

IAM (Identity and Access Management) is an AWS service used to manage users and permissions securely. Created a new IAM user and assigned the required permissions. Set an AWS Budget to monitor cloud expenses and control spending. Configured alerts to track budget limits.

**Steps:**

1. Open the AWS Management Console and navigate to IAM (Identity and Access Management).

2. Select Users, choose Create user, and enter a user name.

3. Configure permissions by attaching the required policy, then review and create the IAM user.

4. Open the created user and go to Security credentials. Create an access key for command-line/programmatic use.

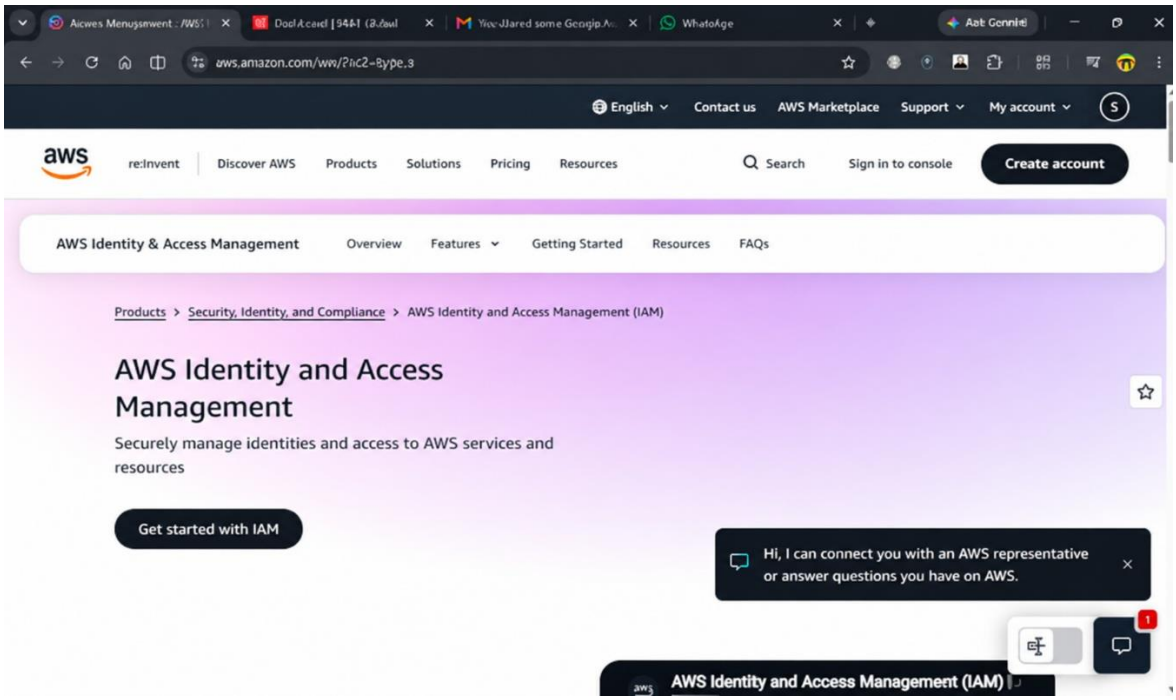
5. Download and securely store the Access Key ID and Secret Access Key. Do not share these credentials.

6. Open Billing and Cost Management, select Budgets, and choose Create budget.

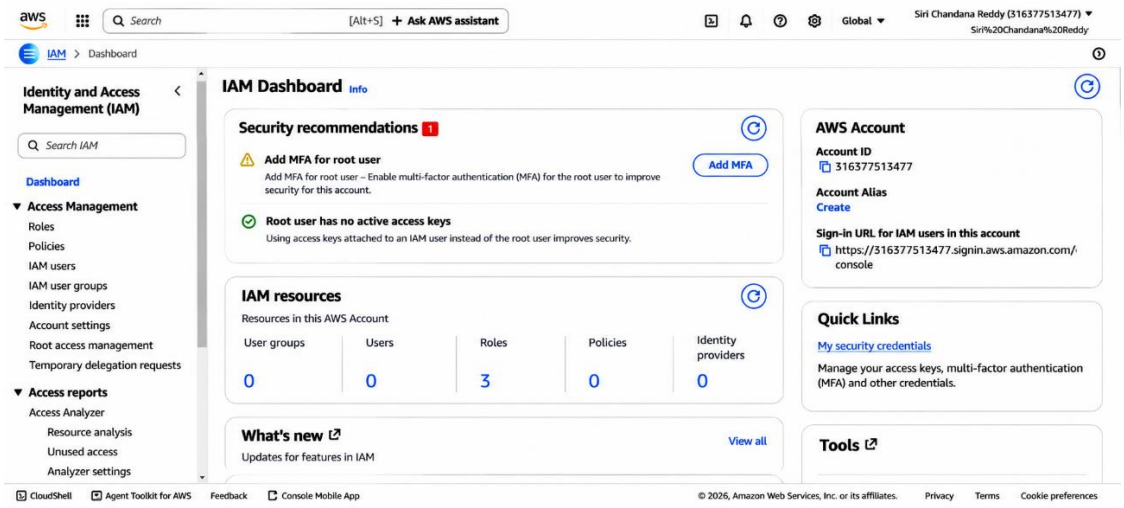
7. Enter a budget name and email recipient, configure the budget threshold, and create the budget to receive email alerts.

**Output:**

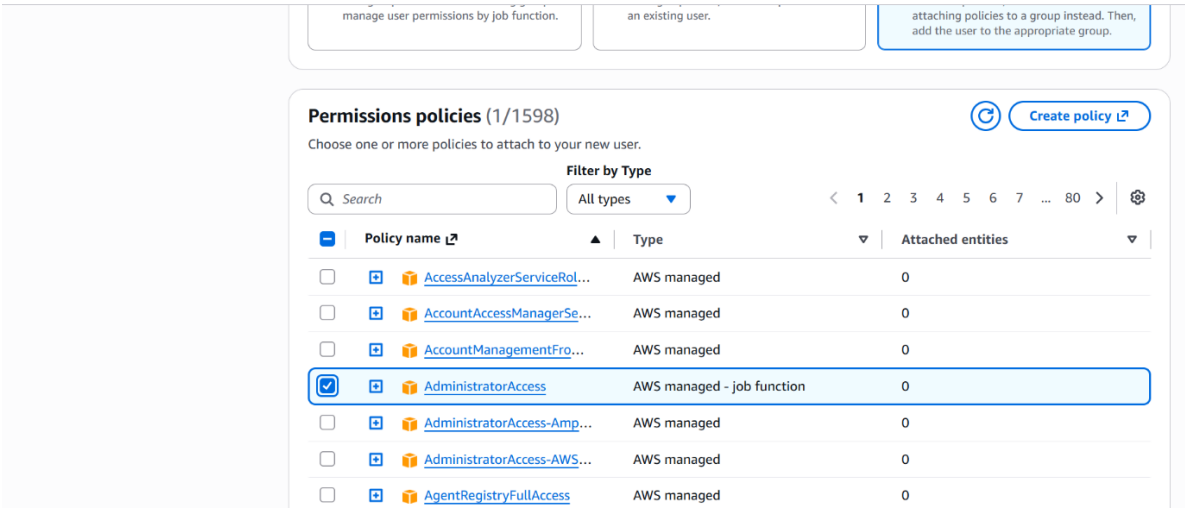
**AWS IAM home page and dashboard**



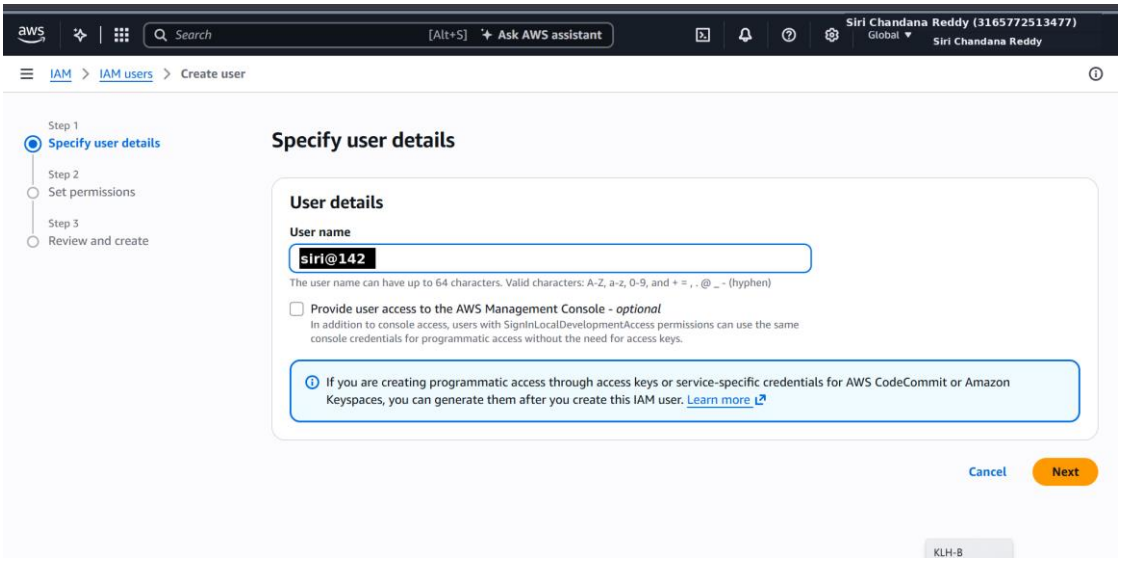

## IAM user creation and permission selection



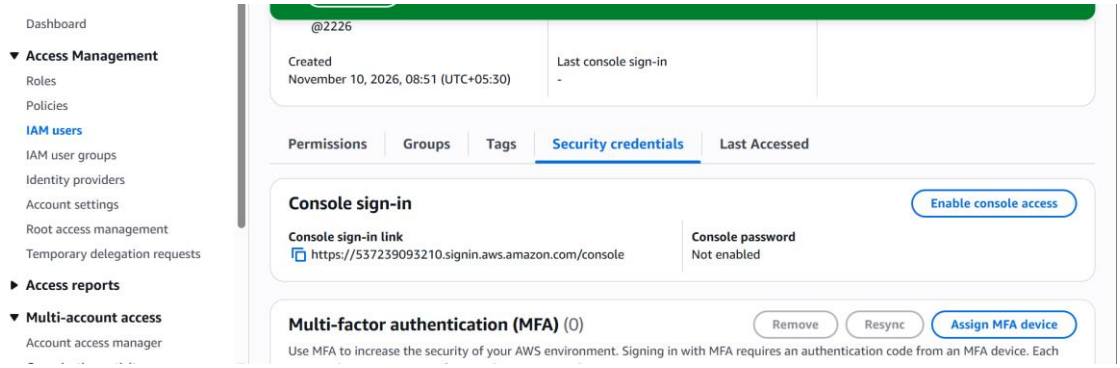

## Created IAM user and user summary



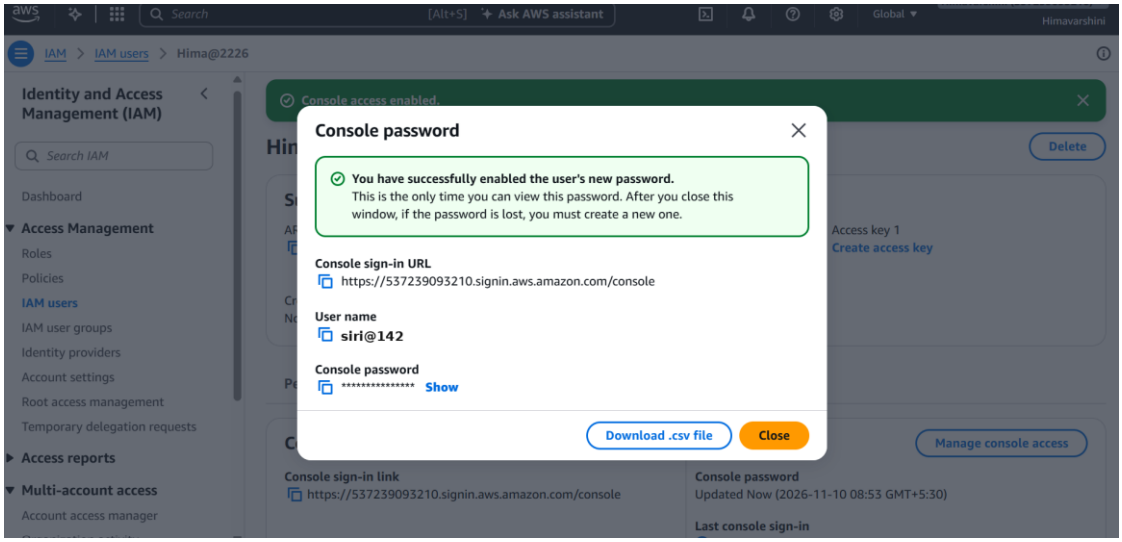

## Security credentials and console access settings

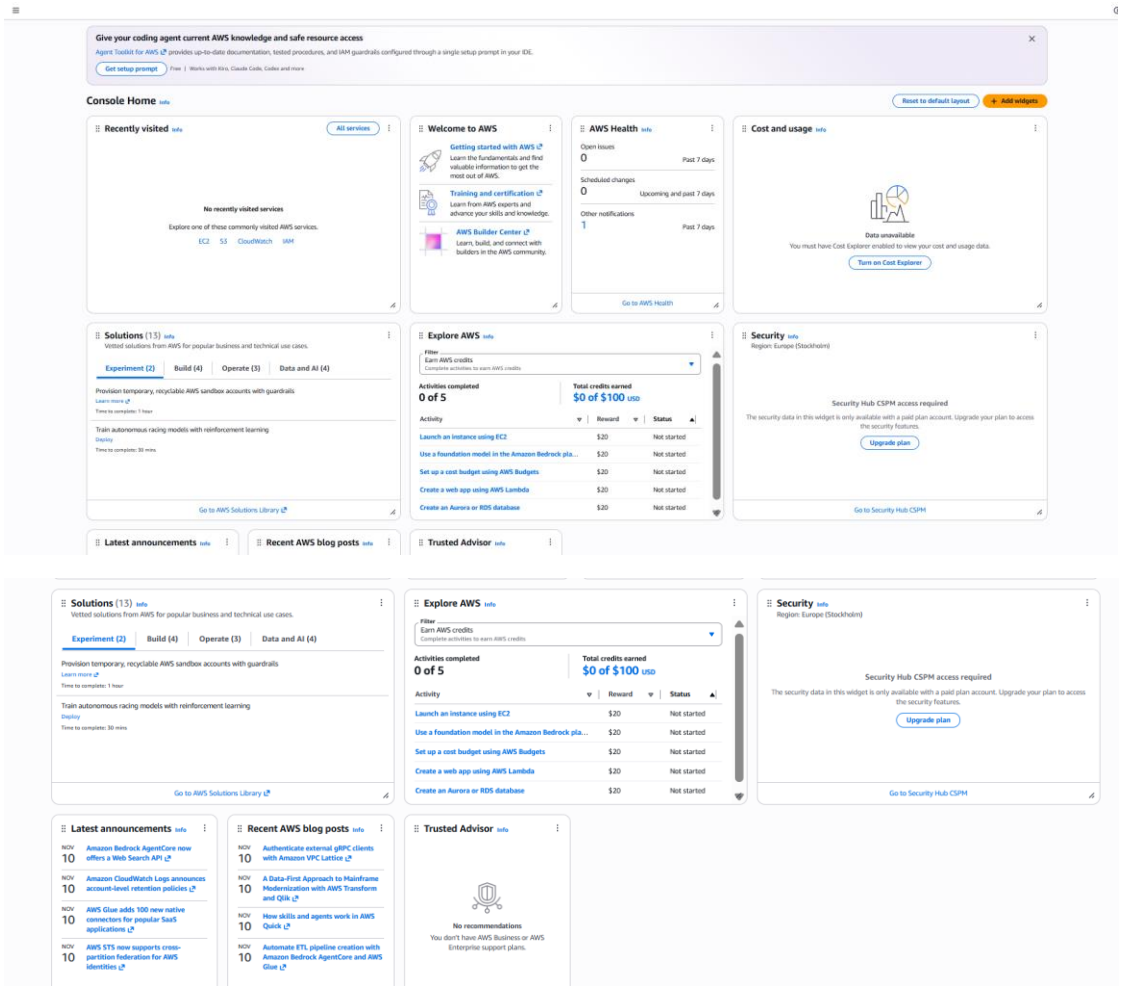


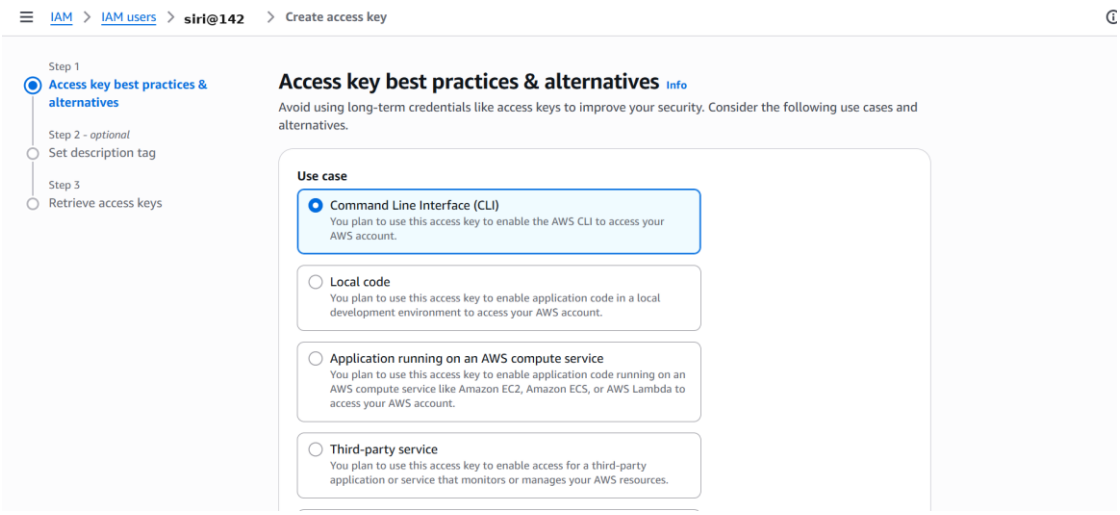

## AWS console home page

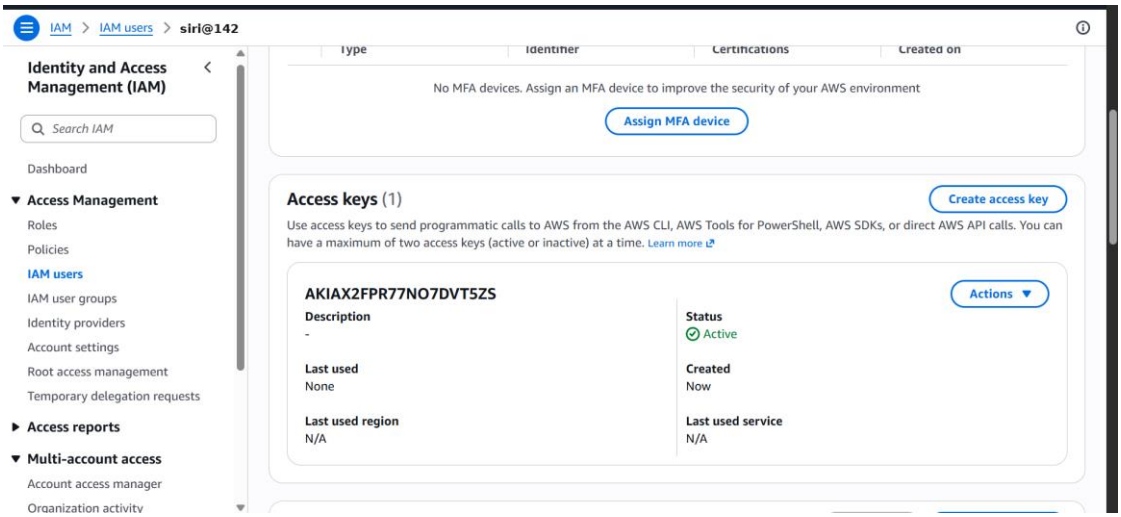


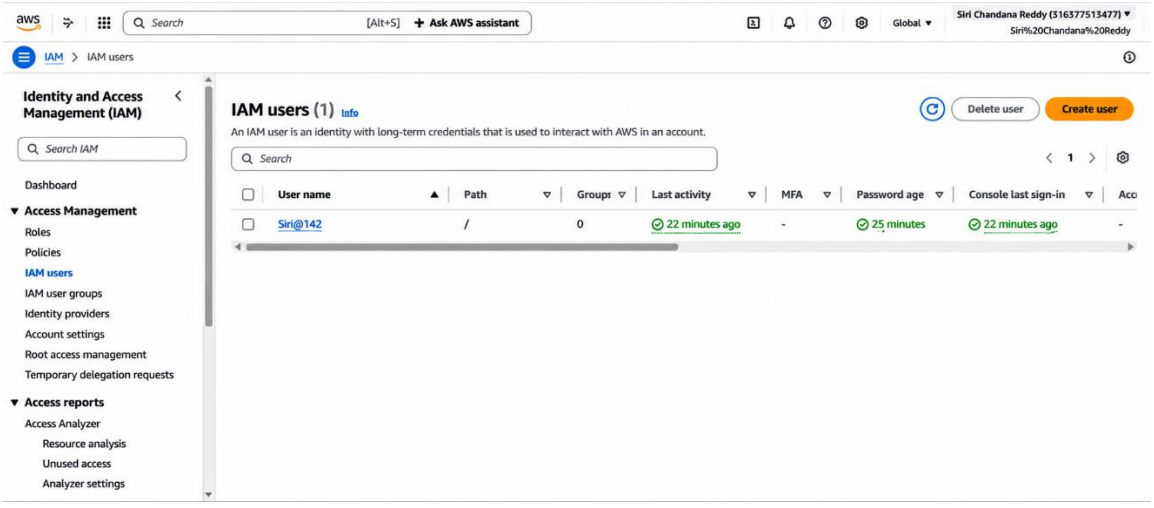
# Access key creation and security credentials

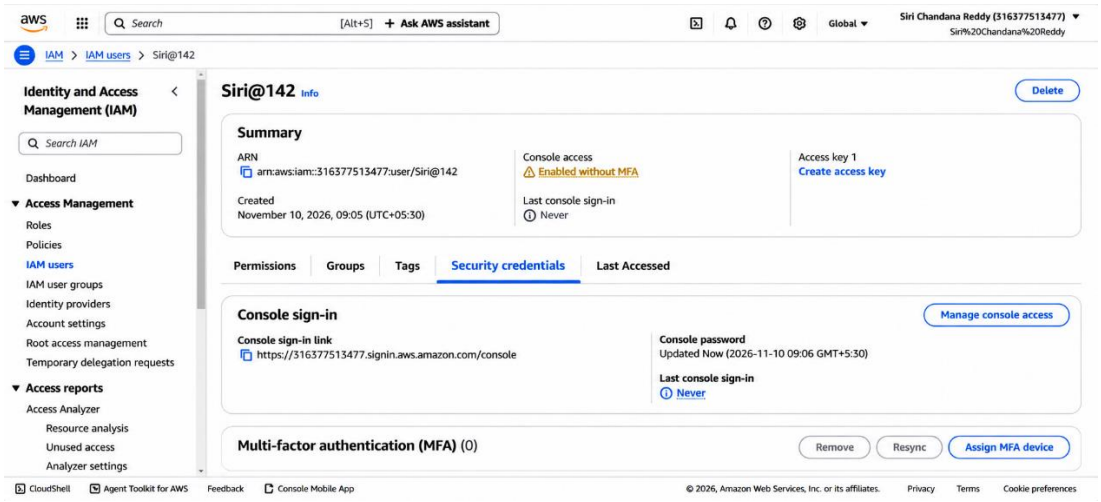


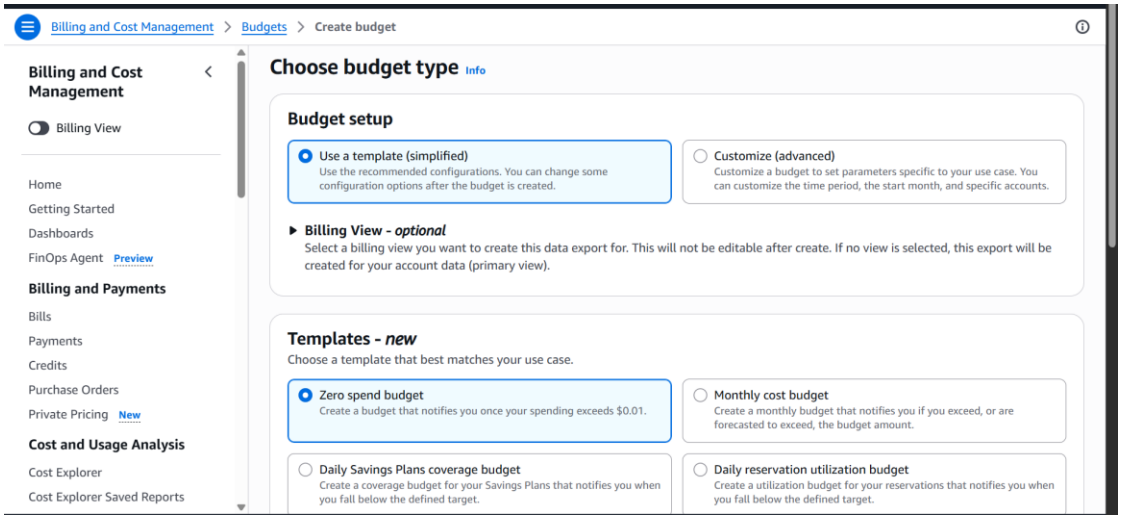




IAM > IAM users > siri@142 > Create access key
Step 1
Access key best practices & alternatives
Step 2 - optional
Set description tag
Step 3
Retrieve access keys
Access key best practices & alternatives Info
Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.
Use case
Command Line Interface (CLI)
You plan to use this access key to enable the AWS CLI to access your AWS account.
Local code
You plan to use this access key to enable application code in a local development environment to access your AWS account.
Application running on an AWS compute service
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.
Third-party service
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.


IAM > IAM users > siri@142
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Multi-account access
Account access manager
Organization activity
Type
Identifier
Certifications
Created on
No MFA devices. Assign an MFA device to improve the security of your AWS environment
Assign MFA device
Access keys (1)
Create access key
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. Learn more
AKIAX2FPR77NO7DVT5ZS
Actions
Description
-
Status
Active
Last used
None
Created
Now
Last used region
N/A
Last used service
N/A


Search
[Alt+S]
Ask AWS assistant
Global
Siri Chandana Reddy (316377513477)
Siri%20Chandana%20Reddy
IAM > IAM users
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
IAM users (1) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete user
Create user
Search
User name
Path
Groups
Last activity
MFA
Password age
Console last sign-in
Siri@142
/
0
22 minutes ago
-
25 minutes
22 minutes ago
-




## Output:

The IAM user was created successfully with programmatic access, and the access credentials were downloaded. The AWS budget was created with an email notification recipient to monitor usage and alert when the configured threshold is reached.

